

Topic 1

Understanding Numbers 0–20

Recognizing, reading, and understanding
what numbers represent.

What are numbers?

Numbers tell us how many
objects we have. We use numbers
every day to count, compare
and describe the world around us.

Each number
represents a
specific amount!

**Numbers from 0 to 20**

0	1	2	3	4	5	6	7	8	9	
10	11	12	13	14	15	16	17	18	19	20

Examples

This is 3.
It means three apples.



This is 7.
It means seven stars.



This is 12.
It means twelve dots.

Topic 2

Counting Forward from 1 to 20

Let's practice saying the numbers in the right order,
from 1 to 20!

When we count forward...

We start at a number and say the
next number, one by one.
Each number is 1 more than
the number before it.

Count slowly
and say each
number out loud!

**Let's count from 1 to 20!**

1

2

3

4

5

6

7

8

9

10

11

12

13

14

15

16

17

18

19

20

Examples**Start at 1**

1, 2, 3, 4, 5

We can count objects
one by one.

Start at 7

7, 8, 9, 10

We can start counting
at any number.

Start at 16

16, 17, 18, 19, 20

Each number is
1 more than the one
before it.

Topic 3

Using a Number Line from 0 to 20

A number line helps us see numbers in order from 0 to 20. Numbers get bigger as we move to the right and smaller as we move to the left.

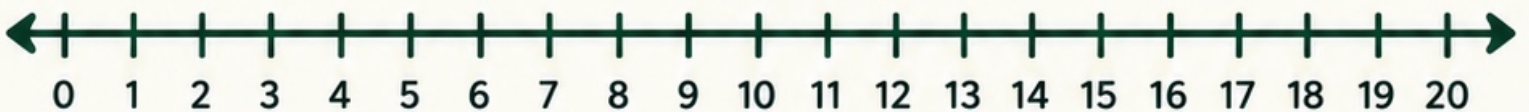
What is a number line?

A number line shows numbers in order. We can use it to count, compare, and find numbers.

Hop right for
bigger numbers!
Hop left for
smaller numbers!



Number line from 0 to 20



What do we notice?

1

Numbers are
in order.

2

Move right to
count forward.

3

Move left to
go backward.

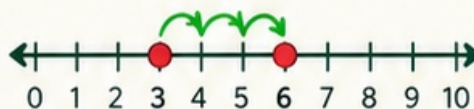
Examples

Find 5



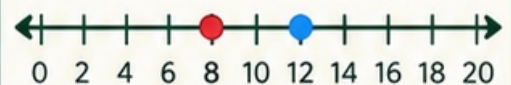
5 is here on the number line.

Hop from 3 to 6



We moved right 3 hops.

Compare 8 and 12



12 is to the right of 8,
so 12 is greater.

Topic 4

Counting Backward from 20 to 0

Let's practice saying the numbers in reverse order,
from 20 down to 0!

When we count backward...

We start at a number and say
the number before it, one by one.
Each number is 1 less than the
number before it.

Count slowly
and go back one
number at
a time!

**Let's count from 20 to 0!**

20	19	18	17	16	15	14	13	12	11	
10	9	8	7	6	5	4	3	2	1	0

Examples**Start at 20****20, 19, 18, 17, 16**

We can count backward
from any number.

Start at 12**12, 11, 10, 9**

Each step backward
is 1 less.

Blast off!**5, 4, 3, 2, 1, 0**

Counting backward helps
us get ready for
subtraction.

Topic 5

Counting Objects with One-to-One Matching

We count by touching or moving each object exactly once.

What does one-to-one matching mean?

When we count, we match one number with one object. We touch or point to each object once and say one number for each object.

Count each
object only once!



How to count objects

- 1 Start at 1.
- 2 Touch or point to each object, one by one.
- 3 Say the numbers out loud.
- 4 Make sure you count each object only once.
- 5 The last number you say tells the total.

Examples

Count the apples.



There are 5 apples.

Count the stars.



There are 4 stars.

Count the cubes.



There are 3 cubes.

Topic 6

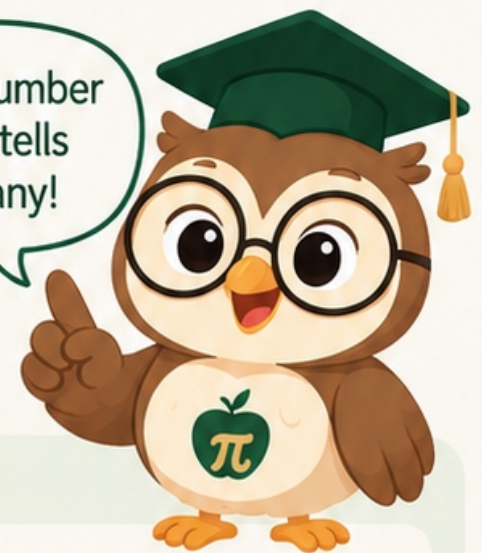
Understanding “How Many?” The Last Number Tells the Total

When we count objects, the last number we say tells how many there are in total. This is called **cardinality**.

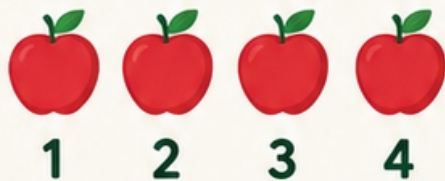
Key Idea

We count each object once, and the last number we say tells the total number of objects.

The last number
you say tells
how many!

**Examples**

Count the apples.



The last number is 4.
There are 4 apples.

Count the stars.



The last number is 5.
There are 5 stars.

Count the cubes.



The last number is 3.
There are 3 cubes.

Your Turn!

Count the objects and write how many there are.



There are ____ circles.



There are ____ triangles.



There are ____ hearts.

Topic 7

Comparing Quantities More, Fewer, and Equal

We can compare two groups of objects and decide which has more, fewer, or the same amount.

Key Words

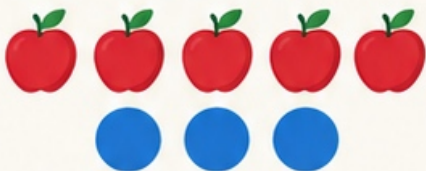
- More** means a greater amount.
(has more objects)
- Fewer** means a smaller amount.
(has fewer objects)
- Equal** means the same amount.
(has the same number of objects)

Look, count
and compare!



We can use our eyes,
our fingers, or count to compare!

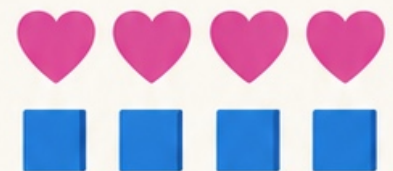
Examples



5 is **more** than 3.
The apples have more
than the circles.



3 is **fewer** than 5.
The stars have fewer
than the triangles.



4 is **equal** to 4.
The hearts and squares
have the same amount.

Your Turn!

Look at each pair. Write more, fewer, or equal.



The stars have _____
than the circles.



The apples have _____
than the triangles.



The hearts and squares are
_____.

Topic 8

Comparing Numbers with <, >, and =

We can use special symbols to compare numbers up to 20.

The Symbols



greater than
means a bigger number

$$8 > 5$$

"8 is greater than 5."



less than
means a smaller number

$$3 < 7$$

"3 is less than 7."



equal to
means the same number

$$6 = 6$$

"6 is equal to 6."

Compare
the numbers
and choose
the correct
symbol!



Look at the numbers.
Think about which one is bigger,
which one is smaller, or if they
are the same.

Examples

$$12 > 7$$



12 is greater than 7.

12 has more apples
than 7.

$$4 < 9$$



4 is less than 9.

4 has fewer stars
than 9.

$$5 = 5$$



5 is equal to 5.

Both groups have
the same amount.

Your Turn!

Write the correct symbol (>, <, or =) in each box.

1. 8 11

2. 6 3

3. 14 18

4. 15 15

5. 20 17

6. 9 9